

SCALABILITY & FUTURE PLAN

Team ID	SWTID-2026-6203
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value

Scalability & Future Plan

This document describes how the Script-Controlled ACL project can be extended to support more users, records, and access requirements. It also outlines possible improvements beyond the current Institution Details table and branch-based access setup.

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	Access rules are demonstrated using a small set of custom roles (bb1–bb4) and a single custom table.	Managing additional roles and tables may require more ACL configuration and careful testing.	Medium
2	Branch-based access is centered on the ECE, EEE, and CSE choices in the Institution Details table.	New branches or more detailed access rules may require updates to conditions and test cases.	Medium
3	The project focuses on record-level READ, CREATE, WRITE, and DELETE access.	More complex requirements, such as field-level restrictions or department-specific policies, are outside the current demonstrated scope.	Low

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Test users and a limited set of roles are used for demonstration.	Test with larger groups of users and role combinations; document expected access for each role.
2	Data Storage	One custom table: u_institution_details, with student and institution-related fields.	Evaluate related tables and reference fields as data grows; keep table design and field definitions consistent.
3	Performance	ACL behavior is checked through functional testing on the project table.	Test list and form access with larger record volumes and review ACL scripts/conditions for unnecessary complexity.
4	Security	Access is controlled through roles, ACL operations, and a branch-related condition.	Add periodic access reviews, negative testing, and documented checks for unauthorized read, create, update, and delete attempts.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Review and test existing READ, CREATE, WRITE, and DELETE ACLs with different role combinations.	Short Term	Confirm access rules behave as intended and record test outcomes.
Phase 2	Extend branch choices and access conditions to cover additional departments or categories, if required.	Medium Term	Support broader institution data and access scenarios.
Phase 3	Add more detailed access requirements, such as field-level controls or additional role-based rules.	Medium Term	Provide more granular control over sensitive information.
Phase 4	Conduct larger-scale testing, periodic access reviews, and update project documentation.	Long Term	Improve maintainability and prepare the solution for wider use.